Edmondson (1999) Psychometric Item Data Simulation & Validation

AI and its Implications for Validity

Updated: October 26, 2026

Contents

```r
# install.packages(c("MASS", "psych", "lavaan"))
library(MASS)
library(psych)
library(lavaan)
```

```
## This is lavaan 0.7-3
## lavaan is FREE software! Please report any bugs.
```

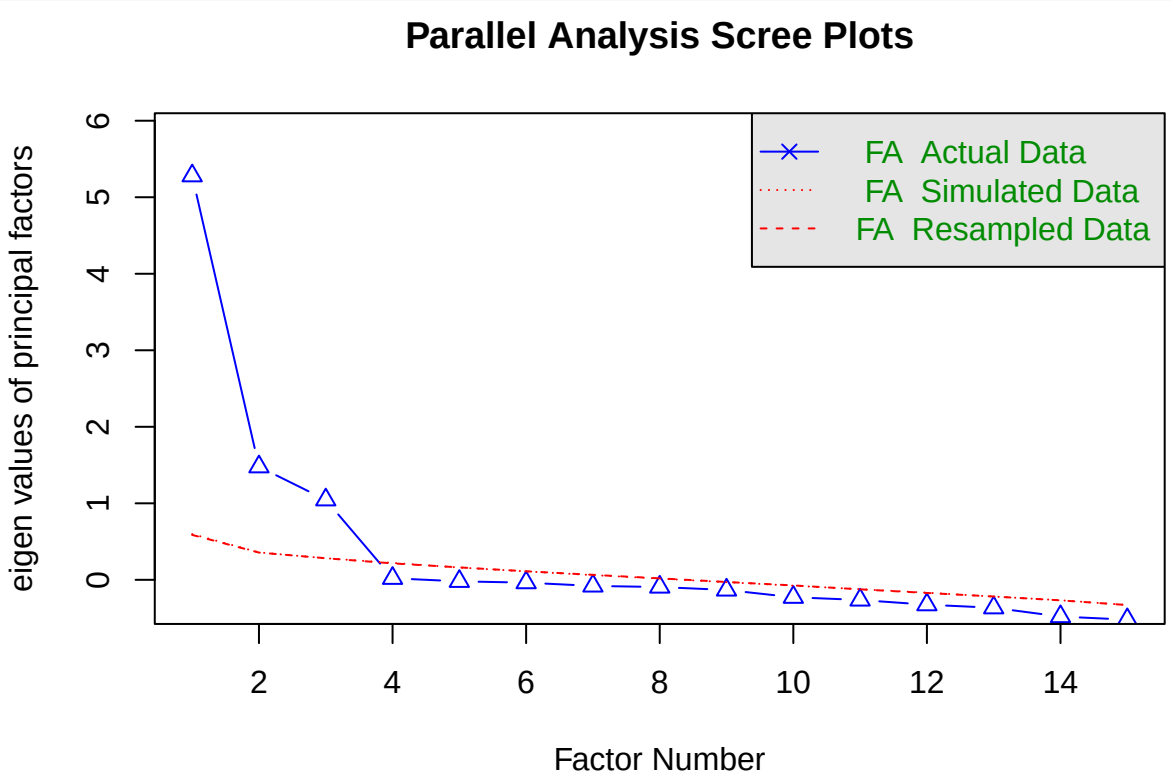
```
##
## Attaching package: 'lavaan'
```

```
## The following object is masked from 'package:psych':
##
##     cor2cov
```

```r
set.seed(123)
n_samples <- 500

# --- Latent factors (TLB = Team Learning Behavior, TP = Team Performance,
#     IM = Internal Motivation, JI = Job Involvement) ---
latent_means <- c(TLB = 4.8, TP = 5.3, IM = 5.6, JI = 4.1)

factor_cov <- matrix(c(
  1.00, 0.42, 0.30, 0.18,
  0.42, 1.00, 0.25, 0.12,
  0.30, 0.25, 1.00, 0.32,
  0.18, 0.12, 0.32, 1.00
), nrow = 4, byrow = TRUE,
dimnames = list(names(latent_means), names(latent_means)))

latent <- as.data.frame(mvrnorm(n_samples, mu = latent_means, Sigma = factor_cov))

# --- Helper: latent score + noise, rounded and bounded to a 1-7 Likert item ---
make_item <- function(latent_score, noise_sd) {
  pmax(1, pmin(7, round(latent_score + rnorm(length(latent_score), 0, noise_sd))))
}

# --- Generate all items (all positively worded, so no reverse scoring needed) ---
my_data <- data.frame(
  # Team Learning Behavior (7 items)
```

```
  sapply(1:7, function(i) make_item(latent$TLB, 0.85)),
  # Team Performance (5 items)
  sapply(c(0.90, 0.85, 0.75, 1.00, 0.90), function(s) make_item(latent$TP, s)),
  # Internal Motivation (3 items)
  sapply(1:3, function(i) make_item(latent$IM, 0.70)),
  # Job Involvement (1 item)
  make_item(latent$JI, 1.15)
)
colnames(my_data) <- c(paste0("TLB_", 1:7), paste0("TP_", 1:5),
                       paste0("IM_", 1:3), "JI_1")

# --- Split into EFA and CFA halves ---
in_efa   <- sample(c(TRUE, FALSE), n_samples, replace = TRUE)
efa_data <- my_data[in_efa, ]
cfa_data <- my_data[!in_efa, ]
```

```
# Exclude single-item JI_1 from EFA
efa_items <- efa_data[, 1:15]

fa.parallel(efa_items, fa = "fa", n.iter = 100)
```




```
## Parallel analysis suggests that the number of factors =  3  and the number of components =  NA
```

```
efa_result <- fa(efa_items, nfactors = 4, rotate = "oblimin", fm = "pa")
```

```
## Loading required namespace: GPArotation
```

```
print(efa_result$loadings, cut = 0.3)
```

```
##
## Loadings:
```

```
##       PA1    PA2    PA3    PA4
## TLB_1  0.737
## TLB_2  0.721
## TLB_3  0.817
## TLB_4  0.771
## TLB_5  0.746
## TLB_6  0.753
## TLB_7  0.819
## TP_1          0.685         0.302
## TP_2          0.727
## TP_3          0.787
## TP_4          0.630
## TP_5          0.698
## IM_1                 0.811
## IM_2                 0.766
## IM_3                 0.770
##
##                    PA1   PA2   PA3   PA4
## SS loadings      4.147 2.526 1.851 0.203
## Proportion Var 0.276 0.168 0.123 0.014
## Cumulative Var 0.276 0.445 0.568 0.582
```

```
cfa_model <- '
  Team_Learning =~ TLB_1 + TLB_2 + TLB_3 + TLB_4 + TLB_5 + TLB_6 + TLB_7
  Performance   =~ TP_1 + TP_2 + TP_3 + TP_4 + TP_5
  Motivation    =~ IM_1 + IM_2 + IM_3
'

cfa_fit <- cfa(cfa_model, data = cfa_data, meanstructure = TRUE)
summary(cfa_fit, fit.measures = TRUE, standardized = TRUE)
```

```
## lavaan 0.7-3 ended normally after 32 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        48
##
##   Number of observations                           243
##
## Model Test User Model:
##
##   Test statistic                                99.788
##   Degrees of freedom                               87
##   P-value (Chi-square)                          0.165
##
##   Browne's residual (NT model-based) test
##   Test statistic                                98.206
##   Degrees of freedom                               87
##   P-value (Chi-square)                          0.193
##
## Model Test Baseline Model:
##
##   Test statistic                              1743.296
##   Degrees of freedom                              105
##   P-value                                       0.000
```

```
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.992
##   Tucker-Lewis Index (TLI)                       0.991
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -5178.522
##   Loglikelihood unrestricted model (H1)      -5128.629
##
##   Akaike (AIC)                               10453.045
##   Bayesian (BIC)                             10620.712
##   Sample-size adjusted Bayesian (SABIC)      10468.559
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.025
##   90 Percent confidence interval - lower         0.000
##   90 Percent confidence interval - upper         0.044
##   P-value H_0: RMSEA <= 0.050                    0.987
##   P-value H_0: RMSEA >= 0.080                    0.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.041
##
## Goodness of Fit Index:
##
##   Goodness of Fit Index (GFI)                    0.994
##   90 Percent confidence interval - lower         0.979
##   90 Percent confidence interval - upper         1.000
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Team_Learning =~
##     TLB_1             1.000                               1.033    0.754
##     TLB_2             1.053    0.084   12.474    0.000    1.087    0.791
##     TLB_3             0.880    0.078   11.343    0.000    0.909    0.726
##     TLB_4             0.939    0.078   11.991    0.000    0.970    0.764
##     TLB_5             0.989    0.082   12.099    0.000    1.022    0.770
##     TLB_6             0.989    0.085   11.656    0.000    1.022    0.744
##     TLB_7             0.978    0.084   11.676    0.000    1.011    0.745
##   Performance =~
##     TP_1              1.000                               0.817    0.696
##     TP_2              1.193    0.117   10.211    0.000    0.974    0.743
##     TP_3              1.147    0.106   10.806    0.000    0.937    0.797
##     TP_4              1.117    0.112    9.948    0.000    0.912    0.721
```

```
##     TP_5              1.159    0.116   10.009    0.000    0.946    0.726
##   Motivation =~
##     IM_1              1.000                              0.857    0.770
##     IM_2              0.899    0.090    9.938    0.000    0.770    0.706
##     IM_3              1.089    0.105   10.396    0.000    0.933    0.825
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Team_Learning ~~
##     Performance       0.283    0.069    4.123    0.000    0.335    0.335
##     Motivation        0.212    0.069    3.050    0.002    0.239    0.239
##   Performance ~~
##     Motivation        0.009    0.053    0.177    0.860    0.014    0.014
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .TLB_1             4.802    0.088   54.619    0.000    4.802    3.504
##    .TLB_2             4.852    0.088   55.051    0.000    4.852    3.532
##    .TLB_3             4.893    0.080   60.897    0.000    4.893    3.907
##    .TLB_4             4.868    0.081   59.767    0.000    4.868    3.834
##    .TLB_5             4.761    0.085   55.915    0.000    4.761    3.587
##    .TLB_6             4.745    0.088   53.866    0.000    4.745    3.455
##    .TLB_7             4.905    0.087   56.391    0.000    4.905    3.617
##    .TP_1              5.362    0.075   71.283    0.000    5.362    4.573
##    .TP_2              5.177    0.084   61.576    0.000    5.177    3.950
##    .TP_3              5.305    0.075   70.377    0.000    5.305    4.515
##    .TP_4              5.346    0.081   65.865    0.000    5.346    4.225
##    .TP_5              5.296    0.084   63.355    0.000    5.296    4.064
##    .IM_1              5.486    0.071   76.871    0.000    5.486    4.931
##    .IM_2              5.494    0.070   78.565    0.000    5.494    5.040
##    .IM_3              5.506    0.073   75.903    0.000    5.506    4.869
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .TLB_1             0.812    0.085    9.498    0.000    0.812    0.432
##    .TLB_2             0.705    0.078    9.074    0.000    0.705    0.374
##    .TLB_3             0.742    0.076    9.732    0.000    0.742    0.473
##    .TLB_4             0.672    0.072    9.400    0.000    0.672    0.417
##    .TLB_5             0.718    0.077    9.334    0.000    0.718    0.407
##    .TLB_6             0.841    0.088    9.584    0.000    0.841    0.446
##    .TLB_7             0.817    0.085    9.574    0.000    0.817    0.444
##    .TP_1              0.708    0.076    9.320    0.000    0.708    0.515
##    .TP_2              0.769    0.088    8.779    0.000    0.769    0.448
##    .TP_3              0.503    0.064    7.842    0.000    0.503    0.364
##    .TP_4              0.768    0.085    9.058    0.000    0.768    0.480
##    .TP_5              0.803    0.089    8.998    0.000    0.803    0.473
##    .IM_1              0.504    0.072    6.966    0.000    0.504    0.407
##    .IM_2              0.596    0.071    8.420    0.000    0.596    0.501
##    .IM_3              0.408    0.076    5.376    0.000    0.408    0.319
##     Team_Learning     1.067    0.160    6.680    0.000    1.000    1.000
##     Performance       0.667    0.114    5.832    0.000    1.000    1.000
##     Motivation        0.734    0.117    6.280    0.000    1.000    1.000
```